# Movable Antenna-Aided MIMO SWIPT Networks

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***Abstract*—This paper investigates a movable antenna (MA)-aided multiple-input multiple-output (MIMO) simultaneous wireless information and power transfer (SWIPT) network, where an access point equipped with multiple MAs simultaneously serves a multi-antenna information decoding user (IDU) and a multi-antenna energy harvesting user (EHU). Our objective is to maximize the harvested power of the EHU by jointly optimizing the transmit precoding (TPC) matrix and MA positions, while satisfying the minimum achievable rate of the IDU. To solve this non-convex problem, we first transform it into an equivalent form by using the weighted minimum mean square error (WMMSE) approach and then propose an efficient alternating optimization algorithm, where the TPC matrix and MA positions are iteratively optimized based on the successive convex approximation (SCA). Simulation results demonstrate that the proposed MA-aided MIMO SWIPT system significantly outperforms conventional fixed-position antenna (FPA) systems and other baselines in terms of energy transfer efficiency.**

## I. Introduction

With the increasing demands on higher transmission rates, multiple-input multiple-output (MIMO) technology has been established as a cornerstone of modern wireless communications [1], [2]. However, conventional MIMO systems with fixed-position antennas (FPAs) may suffer severe performance degradation due to their fixed antenna layouts, lacking the ability to fully exploit continuous channel variations in spatial regions [3]. To overcome this limitation, movable antennas (MAs) and fluid antennas (FAs) have recently emerged as promising technologies, which can exploit additional spatial degrees of freedom (DoFs) by flexibly adjusting the positions of antennas to reshape wireless channels, thereby enhancing communication performance [4], [5].

Recently, MAs/FAs have been widely investigated in various wireless communication scenarios. For instance, the authors in [6] studied MA-enhanced multiuser communications, showing that the total transmit power can be greatly reduced compared to FPA systems via antenna position optimization. Besides, the authors in [7] considered MA-aided multicast communications and proposed a joint transmitter and receiver design to maximize the minimum weighted signal-to-interference-plus-noise ratio (SINR) among all the users. Furthermore, the authors in [8] explored MA-enabled relaying systems and theoretically analyzed the maximum achievable rates for both amplify-and-forward (AF) and decode-and-forward (DF) relays. In addition, MAs have also been applied to other communication scenarios, such as over-the-air computation (AirComp) [9], and integrated sensing and communication (ISAC) [10].

Motivated by these advantages, some research efforts have been devoted to the application of MAs/FAs for enhancing simultaneous wireless information and power transfer (SWIPT) [11]–[13]. However, existing studies are limited to multiple-input single-output (MISO) setups, lacking deeper exploration into MIMO systems to fully exploit the spatial multiplexing gains. To fill this gap, in this paper, we investigate an MA-aided MIMO SWIPT network, where an AP equipped with multiple MAs serves both a multi-FPA information decoding user (IDU) and a multi-FPA energy harvesting user (EHU). Specifically, we formulate an optimization problem to maximize the harvested power of the EHU by jointly optimizing the transmit precoding (TPC) matrix and the positions of all MAs, subject to the minimum achievable rate constraint of the IDU. To tackle the non-convexity of this problem, we first apply the weighted minimum mean square error (WMMSE) approach to transform the original optimization problem and then propose an efficient alternating optimization (AO) algorithm to iteratively optimize the TPC matrix and each MA position based on the successive convex approximation (SCA) technique. Simulation results show that the proposed MA-aided MIMO SWIPT system significantly outperforms conventional FPA systems and other benchmark schemes in terms of the harvested power.

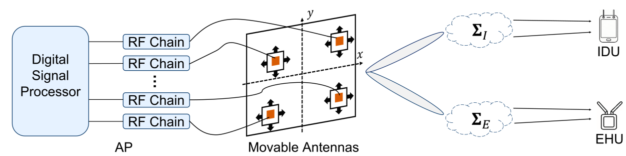


Fig. 1. An MA-aided MIMO SWIPT network.

## II. System Model and Problem Formulation

### A. MA-aided MIMO SWIPT system

As shown in Fig. 1, we consider an MA-aided MIMO SWIPT network. The system comprises an access point (AP) equipped with $M$ MAs, serving an IDU and an EHU simultaneously. The IDU is equipped with $N_I$ FPAs, while the EHU is equipped with $N_E$ FPAs. At the AP, each MA is connected to a radio frequency (RF) chain via a flexible cable, allowing dynamic position adjustment within a two-dimensional (2D) transmit region $\mathcal{C}_t$ of size $A \times A$. The position of the $m$-th MA is denoted by $\boldsymbol{z}_m = [x_m, y_m]^T \in \mathcal{C}_t$.

Let $\boldsymbol{H}_I(\tilde{\boldsymbol{z}})$ and $\boldsymbol{H}_E(\tilde{\boldsymbol{z}})$ denote the channel matrix from the AP to the IDU and the EHU, respectively, with $\tilde{\boldsymbol{z}} = [\boldsymbol{z}_1^T, \cdots, \boldsymbol{z}_M^T]^T$. Then, the received signal at the IDU can be expressed as $\boldsymbol{y}_I = \boldsymbol{H}_I(\tilde{\boldsymbol{z}})\boldsymbol{F}\boldsymbol{s} + \boldsymbol{n}_I$, where $\boldsymbol{s} \in \mathbb{C}^{N_t \times 1}$ represents the information-bearing signal vector for the IDU, with $N_t (1 \leq N_t \leq \min\{M, N_I\})$ denoting the number of data streams and $\mathbb{E}[\boldsymbol{s}\boldsymbol{s}^H] = \boldsymbol{I}_{N_t}$, $\boldsymbol{F} \in \mathbb{C}^{M \times N_t}$ denotes the TPC matrix, and $\boldsymbol{n}_I \sim \mathcal{CN}(\boldsymbol{0}, \sigma_I^2 \boldsymbol{I}_{N_I})$ denotes the additive white Gaussian noise (AWGN) at the IDU. Thus, the achievable rate of the IDU is [14]

$$R = \log\det(\boldsymbol{I}_{N_I} + \boldsymbol{H}_I(\tilde{\boldsymbol{z}})\boldsymbol{F}\boldsymbol{F}^H\boldsymbol{H}_I^H(\tilde{\boldsymbol{z}})\boldsymbol{J}^{-1}), \tag{1}$$

where $\boldsymbol{J} = \sigma_I^2 \boldsymbol{I}_{N_I}$. Similarly, the received signal at the EHU can be expressed as $\boldsymbol{y}_E = \boldsymbol{H}_E(\tilde{\boldsymbol{z}})\boldsymbol{F}\boldsymbol{s} + \boldsymbol{n}_E$, where $\boldsymbol{n}_E \sim \mathcal{CN}(\boldsymbol{0}, \sigma_E^2 \boldsymbol{I}_{N_E})$ denotes the AWGN at the EHU. By neglecting the impact of the noise power, the RF energy harvested by the EHU is given by

$$Q = \eta \text{Tr}(\boldsymbol{H}_E(\tilde{\boldsymbol{z}})\boldsymbol{F}\boldsymbol{F}^H\boldsymbol{H}_E^H(\tilde{\boldsymbol{z}})), \tag{2}$$

where $0 < \eta \leq 1$ denotes the energy harvesting efficiency.

### *B. Field-Response Based Channel Model*

In this paper, we adopt the field-response channel model under the far-field condition [6], where the channel response from the AP to the IDU/EHU is modeled as a superposition of multiple path components. Let $L_v^t, v \in \{I, E\}$ denote the total number of transmit paths from the AP to the IDU/EHU. The elevation and azimuth angles of departure (AoDs) for the $l$-th transmit path between the AP and the IDU/EHU are denoted as $\theta_v^l$ and $\phi_v^l$, respectively. Then, the field response vector (FRV) between the $m$-th MA and the IDU/EHU is

$$\boldsymbol{g}_v(\boldsymbol{z}_m) = [e^{j\frac{2\pi}{\lambda}\rho_v^1(\boldsymbol{z}_m)}, \cdots, e^{j\frac{2\pi}{\lambda}\rho_v^{L_v^t}(\boldsymbol{z}_m)}]^T, \tag{3}$$

with $v \in \{I, E\}$, where $\lambda$ denotes the carrier wavelength and $\rho_v^l(\boldsymbol{z}_m) = (\boldsymbol{\beta}_v^l)^T \boldsymbol{z}_m, 1 \leq l \leq L_v^t$, denotes the signal propagation difference of the $l$-th transmit path between the $m$-th MA's position $\boldsymbol{z}_m$ and the reference point $[0, 0]^T$, with $\boldsymbol{\beta}_v^l = [\sin\theta_v^l \cos\phi_v^l, \cos\theta_v^l]^T$. By stacking the FRVs of all $M$ transmit MAs, we obtain the transmit field response matrix (FRM) for the IDU/EHU as $\boldsymbol{G}_v(\tilde{\boldsymbol{z}}) = [\boldsymbol{g}_v(\boldsymbol{z}_1), \cdots, \boldsymbol{g}_v(\boldsymbol{z}_M)] \in \mathbb{C}^{L_v^t \times M}$. Finally, the channel matrix between the AP and the IDU/EHU can be expressed as

$$\boldsymbol{H}_v(\tilde{\boldsymbol{z}}) = \boldsymbol{R}_v^H \boldsymbol{\Sigma}_v \boldsymbol{G}_v(\tilde{\boldsymbol{z}}), \tag{4}$$

where $\boldsymbol{R}_v \in \mathbb{C}^{L_v^r \times N_v}$ is the receive FRM determined by the fixed geometry of the FPAs at the IDU/EHU, with $L_v^r$ denoting the number of receive paths from the AP to the IDU/EHU, and $\boldsymbol{\Sigma}_v \in \mathbb{C}^{L_v^r \times L_v^t}$ is the path-response matrix from the origin of the transmit region to that of the receive region.

### *C. Problem Formulation*

In this paper, we aim to maximize the power harvested by the EHU for the MA-aided MIMO SWIPT system. To characterize the performance limit of the considered system, we assume that all involved channel state information (CSI) is perfectly known at the AP for system design. Specifically, for the considered system, we aim to maximize (2) by jointly optimizing the precoding matrix $\boldsymbol{F}$ and the positions of MAs $\tilde{\boldsymbol{z}}$. Accordingly, the optimization problem is formulated as:

$$\text{(P1)} \quad \max_{\boldsymbol{F}, \tilde{\boldsymbol{z}}} \quad \eta \text{Tr}(\boldsymbol{H}_E(\tilde{\boldsymbol{z}})\boldsymbol{F}\boldsymbol{F}^H\boldsymbol{H}_E^H(\tilde{\boldsymbol{z}})) \tag{5a}$$

$$\text{s.t.} \quad \|\boldsymbol{F}\|_F^2 \leq P_T, \tag{5b}$$

$$R \geq R^{\min}, \tag{5c}$$

$$\boldsymbol{z}_m \in \mathcal{C}_t, \quad \forall m \in \{1, \cdots, M\}, \tag{5d}$$

$$\|\boldsymbol{z}_m - \boldsymbol{z}_n\|_2 \geq D, \quad \forall n \neq m, \tag{5e}$$

where $P_T$ represents the maximum transmit power budget at the AP, $R^{\min}$ denotes the minimum achievable data rate required by the IDU, and $D$ is the minimum inter-MA distance to avoid mutual coupling effects between each pair of antennas. Problem (P1) is challenging to solve since constraint (5c) involves a complex logarithmic determinant function.

To tackle this issue, we employ the well-known WMMSE method to transform the original problem into an equivalent tractable form [15], which leads to the following optimization problem:

$$\text{(P2)} \max_{\boldsymbol{F}, \tilde{\boldsymbol{z}}, \boldsymbol{U}, \boldsymbol{W}} \eta \text{Tr}(\boldsymbol{H}_E(\tilde{\boldsymbol{z}})\boldsymbol{F}\boldsymbol{F}^H\boldsymbol{H}_E^H(\tilde{\boldsymbol{z}})) \tag{6a}$$

$$\text{s.t.} \quad \log\det(\boldsymbol{W}) - \text{Tr}(\boldsymbol{W}\boldsymbol{E}) + N_t \geq R^{\min}, \tag{6b}$$

$$\text{(5b)}, \text{(5d)}, \text{(5e)},$$

where $\boldsymbol{U}, \boldsymbol{W}$ are two WMMSE auxiliary matrix variables and $\boldsymbol{E} = (\boldsymbol{U}^H\boldsymbol{H}_I(\tilde{\boldsymbol{z}})\boldsymbol{F} - \boldsymbol{I}_{N_t})(\boldsymbol{U}^H\boldsymbol{H}_I(\tilde{\boldsymbol{z}})\boldsymbol{F} - \boldsymbol{I}_{N_t})^H + \sigma_I^2\boldsymbol{U}^H\boldsymbol{U}$. Problem (P2) is still non-convex due to the highly-coupled optimization variables. In the following section, we will propose an AO algorithm to effectively solve this problem.

## III. Proposed AO Algorithm

### *A. Optimizing $\boldsymbol{F}$ for Given $\{\boldsymbol{U}, \boldsymbol{W}, \{\boldsymbol{z}_m\}\}$*

In this subsection, we aim to optimize the TPC matrix $\boldsymbol{F}$ with given $\{\boldsymbol{U}, \boldsymbol{W}, \{\boldsymbol{z}_m\}\}$. The corresponding optimization subproblem can be expressed as

$$\max_{\boldsymbol{F}} \quad \eta \text{Tr}(\boldsymbol{F}^H\boldsymbol{H}_E^H\boldsymbol{H}_E\boldsymbol{F}) \tag{7a}$$

$$\text{s.t.} \quad \text{Tr}(\boldsymbol{F}^H\boldsymbol{F}) \leq P_T, \tag{7b}$$

$$\begin{aligned}&\text{Tr}(\boldsymbol{F}^H\boldsymbol{S}\boldsymbol{F}) - 2\text{Re}\{\text{Tr}(\boldsymbol{W}\boldsymbol{U}^H\boldsymbol{H}_I\boldsymbol{F})\} \leq N_t - R^{\min} \\ &\quad + \log\det(\boldsymbol{W}) - \text{Tr}(\boldsymbol{W}(\boldsymbol{I}_{N_t} + \sigma_I^2\boldsymbol{U}^H\boldsymbol{U})),\end{aligned} \tag{7c}$$

where $\boldsymbol{S} \triangleq \boldsymbol{H}_I^H\boldsymbol{U}\boldsymbol{W}\boldsymbol{U}^H\boldsymbol{H}_I$. Problem (7) still remains non-convex since the objective function in (7a) is non-concave. To address this issue, we employ the SCA technique [16]. Specifically, since (7a) is convex with respect to $\boldsymbol{F}$, we can use the first-order Taylor expansion to construct a linear lower bound on the objective function at the given point $\boldsymbol{F}^{(t)}$, i.e.,

$$\begin{aligned}\eta \text{Tr}(\boldsymbol{F}^H\boldsymbol{H}_E^H\boldsymbol{H}_E\boldsymbol{F}) \geq \eta \text{Re}\{\text{Tr}(2\boldsymbol{F}^{(t)H}\boldsymbol{H}_E^H\boldsymbol{H}_E\boldsymbol{F} \\ - \boldsymbol{F}^{(t)H}\boldsymbol{H}_E^H\boldsymbol{H}_E\boldsymbol{F}^{(t)})\},\end{aligned} \tag{8}$$

where $t$ denotes the iteration index of SCA. Therefore, the optimization problem for $\boldsymbol{F}$ in the $(t+1)$-th iteration can be formulated as

$$\max_{\boldsymbol{F}} \quad \eta \text{Re}\{\text{Tr}(2\boldsymbol{F}^{(t)H}\boldsymbol{H}_E^H\boldsymbol{H}_E\boldsymbol{F} - \boldsymbol{F}^{(t)H}\boldsymbol{H}_E^H\boldsymbol{H}_E\boldsymbol{F}^{(t)})\} \tag{9a}$$

$$\text{s.t.} \quad \text{(7b)}, \text{(7c)},$$

which is a convex quadratically constrained program (QCP) and thus can be solved via the convex solvers [17].

**Algorithm 1** Proposed AO Algorithm for Problem (P2)

Initialize feasible $\boldsymbol{F}^{(0)}$, $\{\boldsymbol{z}_m^{(0)}\}_{m=1}^{M}$; set iteration index $t=0$.
Calculate $\boldsymbol{U}^{(0)}$ and $\boldsymbol{W}^{(0)}$ based on $\boldsymbol{F}^{(0)}$ and $\{\boldsymbol{z}_m^{(0)}\}_{m=1}^{M}$.
**repeat**
    Update the TPC matrix $\boldsymbol{F}^{(t+1)}$ by solving problem (9).
    **for** $m=1$ to $M$ **do**
        Update the MA position $\boldsymbol{z}_m^{(t+1)}$ by solving problem (19).
    **end for**
    Update the auxiliary matrices $\boldsymbol{U}^{(t+1)}$ and $\boldsymbol{W}^{(t+1)}$ based on their closed-form solutions $\boldsymbol{U}^{\star}$ and $\boldsymbol{W}^{\star}$.
    $t \leftarrow t+1$.
**until** The increase of the objective value (6a) is less than $\epsilon$.

### *B. Optimizing $\boldsymbol{z}_m$ for Given $\{\boldsymbol{U},\boldsymbol{W},\boldsymbol{F},\{\boldsymbol{z}_n\}_{n\neq m}\}$*

In this subsection, we aim to alternatively optimize the position of each MA while fixing the other variables, i.e., $\{\boldsymbol{F},\boldsymbol{U},\boldsymbol{W},\{\boldsymbol{z}_n\}_{n\neq m}\}$. Accordingly, the optimization subproblem of the $m$-th MA's position $\boldsymbol{z}_m$ can be expressed as

$$\max_{\boldsymbol{z}_m} \quad \eta \mathrm{Tr}(\boldsymbol{H}_E^H(\boldsymbol{z}_m)\boldsymbol{H}_E(\boldsymbol{z}_m)\boldsymbol{\Phi}) \tag{10a}$$

$$\text{s.t.} \quad \log\det(\boldsymbol{W}) - \mathrm{Tr}(\boldsymbol{W}\boldsymbol{E}(\boldsymbol{z}_m)) + N_t \geq R^{\min}, \tag{10b}$$

$$(5\text{d}), (5\text{e}),$$

where $\boldsymbol{\Phi} \triangleq \boldsymbol{F}\boldsymbol{F}^H$. Problem (10) remains non-convex due to the non-concave objective function (10a) and the non-convex constraints (10b) and (5e). Similarly, we employ the SCA technique to obtain a suboptimal solution.

First, we tackle the non-concave objective function. By substituting the channel matrix $\boldsymbol{H}_E(\tilde{\boldsymbol{z}}) = \boldsymbol{R}_E^H\boldsymbol{\Sigma}_E\boldsymbol{G}_E(\tilde{\boldsymbol{z}})$ into (10a), the objective function can be expanded as

$$\begin{aligned} Q(\boldsymbol{z}_m) &= \eta \mathrm{Tr}(\boldsymbol{H}_E^H(\boldsymbol{z}_m)\boldsymbol{H}_E(\boldsymbol{z}_m)\boldsymbol{\Phi}) \\ &= \eta[\boldsymbol{\Phi}]_{m,m}\boldsymbol{g}_E^H(\boldsymbol{z}_m)\boldsymbol{T}\boldsymbol{g}_E(\boldsymbol{z}_m) + 2\eta\mathrm{Re}\{\boldsymbol{b}^H\boldsymbol{g}_E(\boldsymbol{z}_m)\} \\ &\quad + \eta\sum\nolimits_{n_1\neq m}^{M}\sum\nolimits_{n_2\neq m}^{M}[\boldsymbol{\Phi}]_{n_2,n_1}\boldsymbol{g}_E^H(\boldsymbol{z}_{n_1})\boldsymbol{T}\boldsymbol{g}_E(\boldsymbol{z}_{n_2}), \end{aligned} \tag{11}$$

with $\boldsymbol{T} \triangleq \boldsymbol{\Sigma}_E^H\boldsymbol{R}_E\boldsymbol{R}_E^H\boldsymbol{\Sigma}_E$ and $\boldsymbol{b} \triangleq \sum_{n\neq m}^{M}\boldsymbol{T}^H\boldsymbol{g}_E(\boldsymbol{z}_n)[\boldsymbol{\Phi}]_{m,n}^*$. Let $\nabla Q(\boldsymbol{z}_m)$ and $\nabla^2 Q(\boldsymbol{z}_m)$ denote the gradient vector and Hessian matrix of $Q(\boldsymbol{z}_m)$ over $\boldsymbol{z}_m$, respectively. Then, we can construct a surrogate function to globally lower-bound the objective function by using the second-order Taylor expansion at the given point $\boldsymbol{z}_m^{(t)}$ [18]:

$$\begin{aligned} Q(\boldsymbol{z}_m) &\geq Q(\boldsymbol{z}_m^{(t)}) + \nabla Q^T(\boldsymbol{z}_m^{(t)})(\boldsymbol{z}_m - \boldsymbol{z}_m^{(t)}) \\ &\quad - \frac{\delta_m}{2}(\boldsymbol{z}_m - \boldsymbol{z}_m^{(t)})^T(\boldsymbol{z}_m - \boldsymbol{z}_m^{(t)}) \\ &\triangleq Q_{\text{low}}(\boldsymbol{z}_m), \end{aligned} \tag{12}$$

where $\delta_m$ is a positive parameter that satisfies the condition $\delta_m\boldsymbol{I}_2 \succeq \nabla^2 Q(\boldsymbol{z}_m)$. The detailed derivations for $\nabla Q(\boldsymbol{z}_m)$ and $\nabla^2 Q(\boldsymbol{z}_m)$ are respectively given by

$$\begin{aligned} \nabla Q(\boldsymbol{z}_m) &= -\frac{2\pi\eta}{\lambda}\sum\nolimits_{l_1=1}^{L_E^t}(2|[\boldsymbol{b}]_{l_1}|\sin(\kappa_{l_1}(\boldsymbol{z}_m))\boldsymbol{\beta}_E^{l1} + \\ &\sum\nolimits_{l_2=1}^{L_E^t}[\boldsymbol{\Phi}]_{m,m}|\,[\boldsymbol{T}]_{l_1,l_2}|\sin(\omega_{l_1,l_2}(\boldsymbol{z}_m))(\boldsymbol{\beta}_E^{l_2} - \boldsymbol{\beta}_E^{l_1})), \end{aligned} \tag{13}$$

$$\begin{aligned} \nabla^2 Q(\boldsymbol{z}_m) &= -\frac{4\pi^2\eta}{\lambda^2}\sum\nolimits_{l_1=1}^{L_E^t}(2|[\boldsymbol{b}]_{l_1}|\cos(\kappa_{l_1}(\boldsymbol{z}_m))\boldsymbol{\beta}_E^{l1} \\ &\times(\boldsymbol{\beta}_E^{l1})^T + \sum\nolimits_{l_2=1}^{L_E^t}[\boldsymbol{\Phi}]_{m,m}|\,[\boldsymbol{T}]_{l_1,l_2}|\cos(\omega_{l_1,l_2}(\boldsymbol{z}_m)) \\ &\times(-\boldsymbol{\beta}_E^{l_1} + \boldsymbol{\beta}_E^{l_2})(-\boldsymbol{\beta}_E^{l_1} + \boldsymbol{\beta}_E^{l_2})^T), \end{aligned} \tag{14}$$

where $\omega_{l_1,l_2}(\boldsymbol{z}_m) \triangleq \frac{2\pi}{\lambda}(-\boldsymbol{\beta}_E^{l_1} + \boldsymbol{\beta}_E^{l_2})^T\boldsymbol{z}_m + \arg(\,[\boldsymbol{T}]_{l_1,l_2})$ and $\kappa_{l_1}(\boldsymbol{z_m}) = \frac{2\pi}{\lambda}\boldsymbol{z}_m^T\boldsymbol{\beta}_E^{l_1} - \arg([\boldsymbol{b}]_{l_1})$. In addition, since we have $\|\nabla^2 Q(\boldsymbol{z}_m)\|_F\boldsymbol{I}_2 \succeq \|\nabla^2 Q(\boldsymbol{z}_m)\|_2\boldsymbol{I}_2 \succeq \nabla^2 Q(\boldsymbol{z}_m)$, the value of $\delta_m$ can be set as $\delta_m = \|\nabla^2 Q(\boldsymbol{z}_m)\|_F$.

Next, we aim to address the non-convexity of constraint (10b). By substituting the MSE matrix $\boldsymbol{E}$, the left-handside (LHS) of (10b) can be expanded as

$$\begin{aligned} R(\boldsymbol{z}_m) &= \log\det(\boldsymbol{W}) - \mathrm{Tr}(\boldsymbol{W}\boldsymbol{E}) + N_t \\ &= \log\det(\boldsymbol{W}) + N_t - [\boldsymbol{\Phi}]_{m,m}\boldsymbol{g}_I^H(\boldsymbol{z}_m)\boldsymbol{Y}\boldsymbol{g}_I(\boldsymbol{z}_m) \\ &\quad - 2\mathrm{Re}\{(\sum\nolimits_{n\neq m}^{M}[\boldsymbol{\Phi}]_{m,n}\boldsymbol{g}_I^H(\boldsymbol{z}_n)\boldsymbol{Y} - \boldsymbol{q}_m^T)\boldsymbol{g}_I(\boldsymbol{z}_m)\} \\ &\quad + 2\mathrm{Re}\{\sum\nolimits_{n\neq m}^{M}\boldsymbol{q}_n^T\boldsymbol{g}_I(\boldsymbol{z}_n)\} - \mathrm{Tr}(\boldsymbol{W}(\boldsymbol{I}_{N_t} + \sigma_I^2\boldsymbol{U}^H\boldsymbol{U})) \\ &\quad - \sum\nolimits_{n_1\neq m}^{M}\sum\nolimits_{n_2\neq m}^{M}[\boldsymbol{\Phi}]_{n_2,n_1}\boldsymbol{g}_I^H(\boldsymbol{z}_{n_1})\boldsymbol{Y}_I\boldsymbol{g}_I(\boldsymbol{z}_{n_2}), \end{aligned} \tag{15}$$

where $\boldsymbol{Q} \triangleq \boldsymbol{F}\boldsymbol{W}\boldsymbol{U}^H\boldsymbol{R}_I^H\boldsymbol{\Sigma}_I$, with $\boldsymbol{q}_m^T$ denoting the $m$-th row of $\boldsymbol{Q}$ and $\boldsymbol{Y} \triangleq \boldsymbol{\Sigma}_I^H\boldsymbol{R}_I\boldsymbol{U}\boldsymbol{W}\boldsymbol{U}^H\boldsymbol{R}_I^H\boldsymbol{\Sigma}_I$. Then, we can derive a convex lower bound on $R(\boldsymbol{z}_m)$ at the given point $\boldsymbol{z}_m^{(t)}$:

$$\begin{aligned} R(\boldsymbol{z}_m) &\geq R(\boldsymbol{z}_m^{(t)}) + \nabla R^T(\boldsymbol{z}_m^{(t)})(\boldsymbol{z}_m - \boldsymbol{z}_m^{(t)}) \\ &\quad - \frac{\xi_m}{2}(\boldsymbol{z}_m - \boldsymbol{z}_m^{(t)})^T(\boldsymbol{z}_m - \boldsymbol{z}_m^{(t)}) \\ &\triangleq R_{\text{low}}(\boldsymbol{z}_m), \end{aligned} \tag{16}$$

where $\nabla R(\boldsymbol{z}_m)$ and $\nabla^2 R(\boldsymbol{z}_m)$ denote the gradient and Hessian matrix of $R(\boldsymbol{z}_m)$, respectively; $\xi_m > 0$ is a real parameter satisfying $\xi_m\boldsymbol{I}_2 \succeq \nabla^2 R(\boldsymbol{z}_m)$. The detailed derivations of $\{\nabla R(\boldsymbol{z}_m), \nabla^2 R(\boldsymbol{z}_m), \xi_m\}$ are similar to those of $\{\nabla Q(\boldsymbol{z}_m), \nabla^2 Q(\boldsymbol{z}_m), \delta_m\}$, and are omitted here for brevity. Thus, a convex subset of constraint (10b) is constructed as

$$R_{\text{low}}(\boldsymbol{z}_m) \geq R^{\min}. \tag{17}$$

Finally, the remaining obstacle is to solve the non-convex constraint (5e). According to the Cauchy-Schwarz inequality, we have $(\boldsymbol{z}_m^{(t)} - \boldsymbol{z}_n)^T(\boldsymbol{z}_m - \boldsymbol{z}_n)/\|\boldsymbol{z}_m^{(t)} - \boldsymbol{z}_n\|_2 \leq \|\boldsymbol{z}_m - \boldsymbol{z}_n\|_2$. Thus, a convex subset of constraint (5e) is constructed as

$$(\boldsymbol{z}_m^{(t)} - \boldsymbol{z}_n)^T(\boldsymbol{z}_m - \boldsymbol{z}_n)/\|\boldsymbol{z}_m^{(t)} - \boldsymbol{z}_n\|_2 \geq D, \forall n \neq m. \tag{18}$$

Based on the above, the optimization problem for $\boldsymbol{z}_m$ in the $(t+1)$-th SCA iteration can be formulated as:

$$\max_{\boldsymbol{z}_m} \quad Q_{\text{low}}(\boldsymbol{z}_m) \tag{19a}$$

$$\text{s.t.} \quad (5\text{d}), (17), (18).$$

Problem (19) is a quadratic programming (QP) problem, which can be efficiently solved by using the convex solvers [17].

### *C. Optimizing $\boldsymbol{U}$ and $\boldsymbol{W}$ for Given $\{\boldsymbol{F},\{\boldsymbol{z}_m\}\}$*

In this subsection, we optimize the auxiliary variables $\boldsymbol{U}$ and $\boldsymbol{W}$ with fixed $\{\boldsymbol{F},\{\boldsymbol{z}_m\}\}$. Note that for given TPC matrix $\boldsymbol{F}$ and MA positions $\tilde{\boldsymbol{z}}$, the optimal solution to $\boldsymbol{U}$ and $\boldsymbol{W}$ can

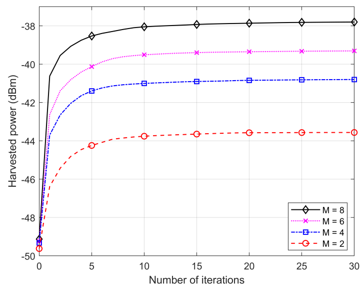


Fig. 2. Convergence behavior of Algorithm 1.

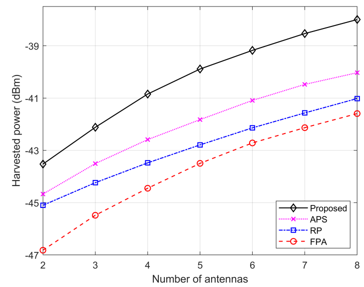


Fig. 3. Harvested power versus $M$.

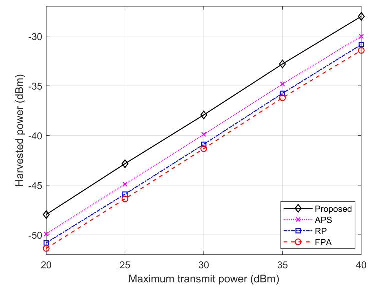


Fig. 4. Harvested power versus $P_T$.

be derived in closed forms by setting the first-order derivatives of the LHS of (6b) with respect to $\boldsymbol{U}$ and $\boldsymbol{W}$ to zero, which yields $\boldsymbol{U}^\star = (\boldsymbol{H}_I\boldsymbol{F}\boldsymbol{F}^H\boldsymbol{H}_I^H + \boldsymbol{J})^{-1}\boldsymbol{H}_I\boldsymbol{F}$ and $\boldsymbol{W}^\star = (\boldsymbol{I}_{N_t} - \boldsymbol{F}^H\boldsymbol{H}_I^H(\boldsymbol{J} + \boldsymbol{H}_I\boldsymbol{F}\boldsymbol{F}^H\boldsymbol{H}_I^H)^{-1}\boldsymbol{H}_I\boldsymbol{F})^{-1}$ [14].

### *D. Overall Algorithm*

The overall AO-based algorithm for solving problem (P3) is summarized in Algorithm 1. In lines 4-9, the TPC matrix $\boldsymbol{F}$, each MA position $\boldsymbol{z}_m$, and auxiliary matrix $\boldsymbol{U}, \boldsymbol{W}$ are alternately optimized. Since the objective value of problem (P2) is non-decreasing over iterations and the solution set is compact due to the finite transmit power, the convergence of Algorithm 1 is guaranteed. Moreover, the computational complexity of Algorithm 1 is primarily dominated by optimizing the TPC matrix and MA positions. Specifically, the complexity of updating the TPC matrix $\boldsymbol{F}$ in line 4 is $\mathcal{O}\big(\ln(\frac{1}{\beta})(MN_t)^3\big)$, with $\beta$ denoting the solution accuracy; the complexity of updating each MA's position in line 6 is $\mathcal{O}\big(\ln(\frac{1}{\beta})M^{1.5} + L_{\max}^2\big)$, with $L_{\max} = \max\{L_I^t, L_E^t\}$. Thus, the complexity of Algorithm 1 is given by $\mathcal{O}\big(T_{\max}(\ln(\frac{1}{\beta})(MN_t)^3 + \ln(\frac{1}{\beta})M^{2.5} + ML_{\max}^2)\big)$, where $T_{\max}$ denotes the maximum number of iterations.

## IV. Numerical Results

In this section, we present simulation results to evaluate the performance of our proposed MA-aided MIMO SWIPT system. The distances from the AP to the IDU and EHU are randomly generated within $[10, 20]$ meters (m) and $[5, 10]$ m, respectively. We adopt the geometry channel model with an identical number of transmit and receive channel paths for the IDU and EHU, i.e., $L_v^t = L_v^r \triangleq L = 5, v \in \{I, E\}$. The path-response matrix for the IDU/EHU is modeled as a diagonal matrix [6], given by $\boldsymbol{\Sigma}_v = \mathrm{diag}(\psi_{v,1}, \cdots, \psi_{v,L})$ with $\psi_{v,l} \sim \mathcal{CN}(0, C_0 d_v^{-a}/L), 1 \le l \le L$, where $d_v$ denotes the distance from the AP to the IDU/EHU, $C_0 = -30$ dB denotes the channel power gain at the reference distance of 1 m, and $a = 2.7$ is the path loss factor. In addition, we set $N_I = N_E = 2$, $N_t = 2$, $D = \lambda/2$, $\sigma_I^2 = -80$ dBm, $A = 4\lambda$, $R^{\min} = 2$ bit/s/Hz, and $\epsilon = 10^{-3}$. All results are averaged over 100 independent trials.

We first evaluate the convergence performance of the proposed Algorithm 1 under different numbers of MAs in Fig. 2, with $P_T = 30$ dBm. We can observe that the harvested power increases monotonically with the iteration index and converges rapidly within 20 iterations for all considered values of $M$.

Next, the performance of Algorithm 1 is compared with the following benchmark schemes: 1) **FPA**: The AP is equipped with a uniform planar array consisting of $M$ FPAs with each spaced by $\lambda/2$; 2) **Random position (RP)**: We randomly generate 500 candidate solutions of $\tilde{\boldsymbol{z}}$ and then select the one that satisfies constraint (6b) and yields the maximum harvested power to replace the antenna position optimization in step 6 of Algorithm 1; 3) **Alternating position selection (APS)**: The total MA moving region at the AP is quantized into discrete locations with distance $\lambda/2$, and the best-performing position of each MA is alternately selected to replace the antenna position optimization in step 6 of Algorithm 1.

Fig. 3 illustrates the EHU's harvested power versus the number of antennas at the AP, i.e., $M$, with $P_T = 30$ dBm. The harvested power of all considered schemes increases with $M$ due to the enhanced spatial diversity and multiplexing gains. Moreover, the proposed approach achieves significantly higher harvested power than the conventional FPA scheme. This is attributed to the flexibility of MAs in antenna repositioning to fully exploit channel spatial variations for creating more favorable channel conditions. Furthermore, the proposed scheme also exhibits superior performance over other benchmark schemes.

Fig. 4 plots the EHU's harvested power versus the maximum transmit power at the AP, i.e., $P_T$, with $M = 8$. It is observed that the harvested power achieved by all schemes increases with the maximum transmit power. In addition, the proposed scheme achieves higher harvested power over other benchmark schemes under any value of $P_T$. In particular, when $P_T = 30$ dBm, the proposed scheme yields $118.5\%$, $97.5\%$, and $56.9\%$ performance gains over the FPA, RP, and APS schemes, respectively, validating the effectiveness of the antenna position optimization in Algorithm 1.

## V. Conclusion

This paper investigated a harvested power maximization problem in an MA-aided MIMO SWIPT system, where the transmit precoding matrix and MA positions were jointly optimized. To address the non-convexity of this problem, we equivalently transformed it by using the WMMSE approach and proposed an efficient algorithm based on AO and SCA techniques. Numerical results demonstrated that the proposed system significantly outperformed traditional FPA systems and other benchmark schemes in terms of the harvested power.

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